

# Advances in the pathophysiology, diagnosis and treatment of Takayasu arteritis

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## Abstract

Takayasu arteritis (TAK) is a rare, chronic, large-vessel vasculitis that primarily targets the aorta and its major branches, leading to vascular stenosis, occlusion and aneurysm formation. TAK, which is characterized by granulomatous inflammation of the arterial wall, predominantly affects women, with peak onset typically occurring between 20 and 40 years of age. The disease exhibits substantial geographic variability in prevalence, with emerging evidence suggesting that these differences are partly owing to variations in genetic susceptibility loci, particularly within immune-related genes; however, the role of environmental factors in the disease aetiology remains poorly understood. Non-invasive imaging techniques have become central to both diagnosis and disease monitoring. Furthermore, the development of biomarkers holds promise for more accurate assessment of disease activity. The management of TAK is evolving, driven by an improved understanding of disease pathogenesis. The growing use of biologic agents is providing new treatment options, particularly for patients with refractory or relapsing disease. By integrating these developments, this Review is aimed at serving as a comprehensive resource for clinicians and researchers dedicated to improving the understanding and management of TAK.

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## Key points

- Takayasu arteritis (TAK) is a rare large-vessel vasculitis with a global incidence of 1.11 per million person-years, which predominantly affects young women 20–40 years of age.
- The pathogenesis of TAK involves multiple interconnected immune-mediated processes that ultimately result in vascular fibrosis and stenosis.
- Over 100 genetic susceptibility loci, including *HLA-B\*52*, have been identified, contributing to geographic differences in disease prevalence.
- Non-invasive imaging is central to diagnosis and monitoring;  $^{18}\text{F}$ -fluorodeoxyglucose-PET uptake in the arterial wall predicts future stenotic lesions better than wall thickness alone.
- Serum pentraxin-3 shows promise as a biomarker of disease activity, particularly when IL-6 blockers are used.
- Glucocorticoids combined with conventional immunosuppressive therapy remain the cornerstone of treatment; relapse rates with glucocorticoid monotherapy reach 60–77% within 1 year.
- Biologic agents, including TNF inhibitors and tocilizumab, can be effective in refractory TAK; therapies such as secukinumab and Janus kinase inhibitors are emerging options.

## Introduction

Takayasu arteritis (hereafter also referred to as TAK), first described by the Japanese ophthalmologist Mikito Takayasu in 1908, is characterized by chronic granulomatous inflammation of the arterial wall that can lead to stenosis, occlusion and aneurysm formation<sup>1</sup>. These vascular changes can result in ischaemic symptoms and the characteristic loss of pulses, hence why TAK is sometimes referred to as ‘pulseless disease’. Disruption of blood flow through crucial arteries to the brain, heart and other organs can be life threatening.

TAK is more common in women than in men; reported ratios vary between 2:1 and 9:1 (ref. 2). Although disease onset is typically 20–40 years of age, childhood-onset disease and onset after 40 years of age can occur<sup>3</sup>. TAK is a rare disease; the estimated global incidence of TAK is 1.11 per million person-years, although incidence rates vary considerably among different populations<sup>4</sup>. The disease is most common in East Asian populations, with the highest prevalence rates reported in Japan (40 per million)<sup>5</sup>. However, the disease can affect individuals of any genetic ancestry. Studies suggest that inter-population differences in disease prevalence are, at least in part, caused by variation in allele frequencies at disease-associated genetic susceptibility loci<sup>6</sup>. The role of environmental factors in the aetiology of TAK is less clear.

Given its complex and often insidious onset, TAK requires a high index of clinical suspicion and a multidisciplinary approach to diagnosis and management. In this Review, we comprehensively examine advances in the pathophysiology, clinical features, diagnostic strategies and therapeutic approaches with regard to TAK.

## Pathology and immunopathogenesis

In TAK, the arterial tissue has a granulomatous pan-arteritis with prominent adventitial fibrosis, neovascularization, disruption of the elastic laminae and intima-media thickening (Fig. 1). Environmental factors, particularly infectious triggers, are thought to contribute to disease development. This section explores the immunopathological mechanisms underlying TAK, including the roles of microbial exposure, dysregulated innate and adaptive immune-cell activation, cytokine signalling and tissue remodelling, which lead to arterial fibrosis.

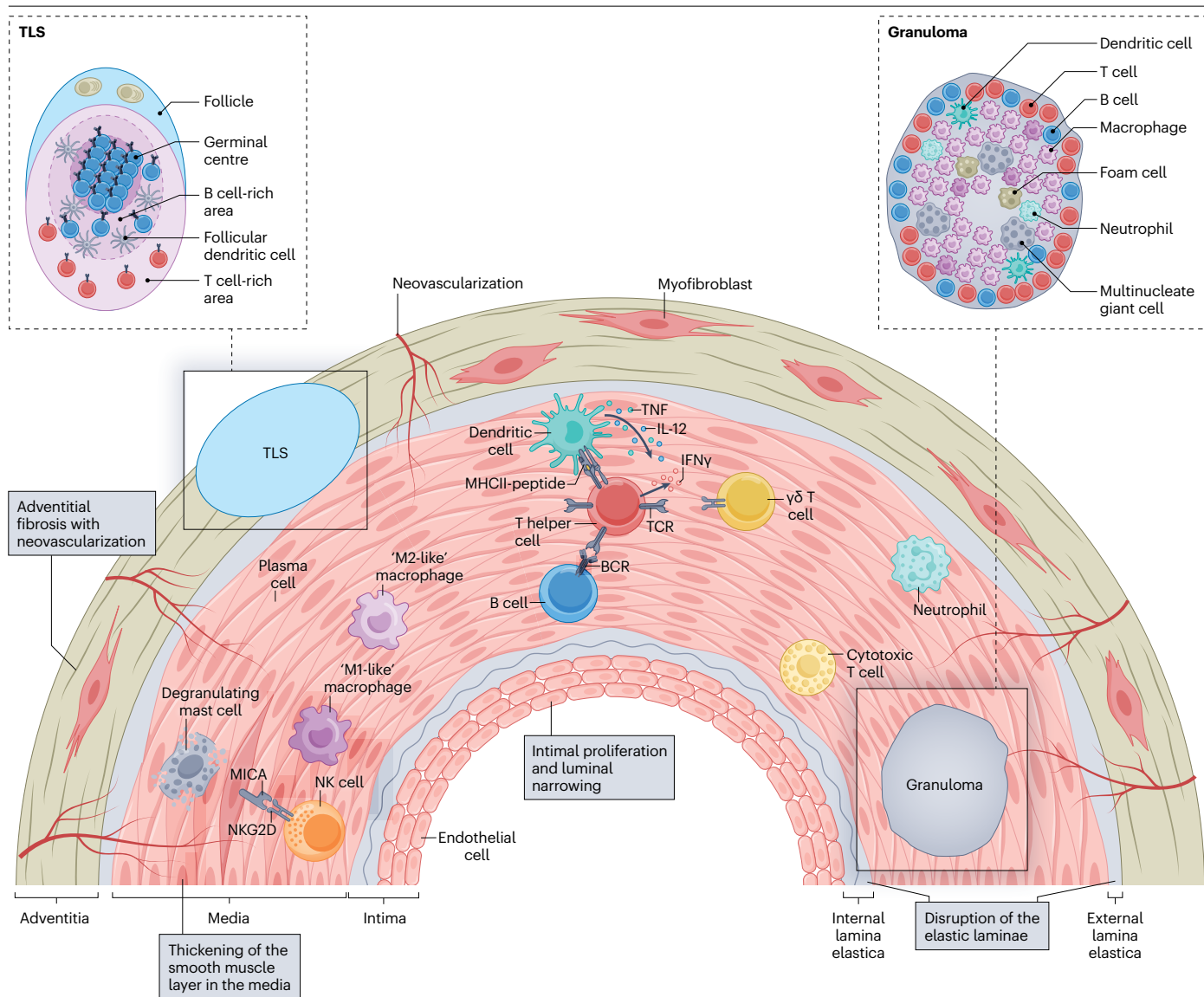
### Environmental factors

Indirect evidence suggests an association between TAK and infections; for example, a high frequency of tuberculosis occurs in patients with TAK (including those not on immunosuppressive therapy)<sup>7,8</sup>. In addition, the common pathology of granulomatous inflammation, reports of primary immunodeficiency disorders associated with paediatric-onset TAK and the role of T helper 17 (T<sub>H</sub>17) cells in TAK (a cell type that is linked to viral infections) make it reasonable to hypothesize that an aberrant immune response to prior infection with tuberculosis, viruses or other infections might trigger TAK<sup>7–12</sup>. Compared with atherosclerotic arteries, the arterial wall is inflamed in TAK and shows increased MHC class I polypeptide-related sequence A (MICA) expression, which can be induced by infections and can activate infiltrating  $\gamma\delta$  T cells and natural-killer (NK) cells via the NKG2D receptor<sup>13–20</sup>. In TAK, a greater abundance of *Streptococcus* and *Campylobacter* species has been reported in the gut microbiota, and in the blood, Clostridia, Cytophagia and Deltaproteobacteria are increased, whereas Bacilli classes are reduced compared with healthy individuals<sup>21,22</sup>.

### Innate immunity

The activation of  $\gamma\delta$  T cells and NK cells in TAK arterial tissue is hypothesized to stimulate their cytotoxic activity and is also associated with higher expression of the co-stimulatory molecules CD137 (also known as 4-1BB) and Fas in the arterial wall, which in turn enhance T cell activation<sup>19,20</sup> (Fig. 1). Vascular smooth-muscle cells in patients with TAK have a senescent phenotype with mitochondrial dysfunction, driven by IL-6 (ref. 23).

The gene *ETS2*, which regulates monocyte and macrophage activation, has been identified as a common genetic association between TAK, ankylosing spondylitis and inflammatory bowel diseases, which are also linked through their shared MHC-I associations (referred to as MHC-I-opathies)<sup>24,25</sup>. However, clinical and molecular characteristics suggest that TAK belongs to a distinct subset of MHC-I-opathies<sup>25</sup>, demonstrated by the lack of an association with ERAP1 or ERAP2 in contrast with other MHC-I-opathies (A.H.S., unpublished observations). A polymorphism (rs2069837) in the *IL6* enhancer region that alters the expression of glycoprotein non-metastatic melanoma protein B (GPNMB), a negative regulator of monocyte and macrophage activation, has been linked with TAK susceptibility<sup>26</sup>. Keeping in mind the continuum between M1-like and M2-like macrophage polarization within biological systems, patients with TAK who are not on immunosuppressive therapy show a predominant infiltration of M1-like macrophages and the chemoattractant protein CCL2 is detected in the adventitia, whereas those who are on immunosuppressants have primarily M2-like macrophage infiltration and CCL2 reported in the media layer<sup>27</sup>. These findings indicate a shift to a predominantly fibrotic M2-like phenotype in patients with TAK receiving immunosuppressive therapy, which corresponds



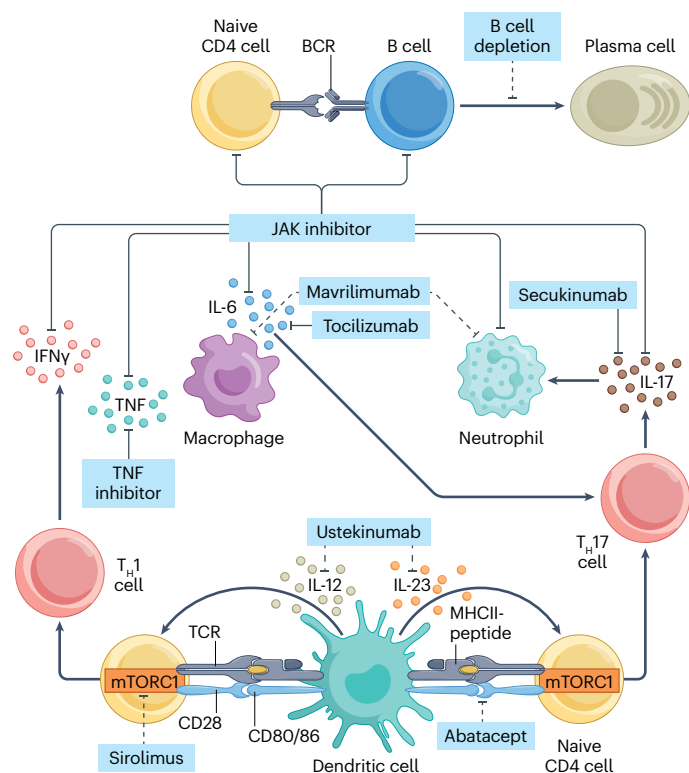
**Fig. 1 | Pathology of Takayasu arteritis.** The pathology of Takayasu arteritis (TAK) is a pan-arteritis, believed to begin in the adventitial layer, initially through vasa vasorum affliction, before involving the media and intima, resulting in intimal hyperplasia, disruption of the elastic laminae and adventitial thickening with fibrosis and neovascularization<sup>13,14</sup>. Cellular infiltrates in the arterial wall include T cells, B cells, plasma cells,  $\gamma\delta$  T cells, natural killer (NK) cells, dendritic cells, neutrophils and macrophages<sup>13,15</sup>. Granulomatous arteritis is common to TAK and giant-cell arteritis (GCA). Granulomas with multinucleate giant cells in

the TAK arterial wall are often found in the intima and media. The arteries from patients with TAK show a dominant infiltration of CD8<sup>+</sup> T cells and NK cells<sup>14</sup>. Tertiary lymphoid structures (TLS) have been identified in the adventitia of TAK arteries<sup>16,17</sup>. Arterial-wall fibrosis (which results in arterial stenosis) is prominent in TAK, unlike GCA, in which aneurysms are more common<sup>18</sup>. BCR, B cell receptor; MICA, MHC class I polypeptide-related sequence A; TCR, T cell receptor. The original version of this figure was created with BioRender.com.

with the stenosing behaviour of arteritis in TAK<sup>18</sup>. Patients with active TAK have higher frequencies of circulating total monocytes and classical and intermediate sub-populations than healthy people, and levels of the monocyte chemokine CCL22 are elevated compared with those with inactive TAK<sup>28</sup>. Arterial-wall macrophage infiltration detection in vivo using somatostatin-2 receptor PET, which targets the somatostatin-2 receptor expressed on inflammatory macrophages, merits evaluation as a novel disease activity measure for TAK<sup>29</sup>.

### Adaptive immunity

Dendritic cells infiltrating affected arteries might bridge innate and adaptive immunity in TAK<sup>13</sup>.  $T_H1$  and  $T_H17$  cells, via secretion of IFN $\gamma$  and IL-17, respectively, drive granuloma formation and arteritis in TAK<sup>10,25,30</sup>. In TAK, the activation of mammalian target organ of rapamycin (mTOR), particularly mTOR complex 1 (mTORC1) in CD4<sup>+</sup> T cells is driven by Notch-1, which promotes  $T_H1$  and  $T_H17$  cell activation<sup>31,32</sup>. Unlike in giant-cell arteritis (GCA), another form of large-vessel



**Fig. 2 | Therapeutic targets in Takayasu arteritis.** Observational data and the secondary end point of a randomized controlled trial (RCT) support the use of IL-6 inhibition with tocilizumab in Takayasu arteritis (TAK)<sup>40,41</sup>. The use of TNF inhibitors and secukinumab for TAK is also supported in observational studies<sup>41,42</sup>. Costimulatory blockade with abatacept and IL12p40 inhibition with ustekinumab failed to demonstrate benefits versus placebo in RCTs, although methodological issues and premature termination rendered the trial of ustekinumab at a high risk of bias<sup>43–45</sup>. Despite literature supporting the role of B cells in the arterial pathology of TAK, observational data reveal the ineffectiveness of rituximab in TAK, although other B cell-depleting therapies, such as obinutuzumab and inebilizumab, remain unexplored<sup>46</sup>. Sirolimus blocks mTORC1, which drives T helper 1 (T<sub>H</sub>1) cell and T<sub>H</sub>17 cell differentiation in TAK, and is a potential treatment target for future exploration<sup>32</sup>. Mavrilimumab blocks granulocyte-macrophage colony-stimulating factor receptor which activates macrophages and neutrophils<sup>201</sup>; it has been proven effective in an RCT of giant-cell arteritis, although it remains unexplored in TAK<sup>202</sup>. Observational studies have reported the effectiveness of Janus kinase (JAK) inhibitors in TAK and these drugs target multiple cytokines and immune cells. The solid lines indicate therapies that are commonly used with some evidence to support their use in patients with TAK. The dotted lines indicate potential future therapies based on an understanding of the pathogenesis of TAK or therapies that have a mechanistic basis but have not yet been proven to be effective in patients with TAK. CD80/86, CD80 or CD86; mTORC1, mammalian target organ of rapamycin complex 1; TCR, T cell receptor; BCR, B cell receptor. The original version of this figure was created with BioRender.com.

vasculitis, the expansion of T<sub>H</sub>17 cells in the peripheral blood in TAK is not inhibited by glucocorticoids<sup>10,30</sup>. T<sub>H</sub>17.1 cells have been reported in TAK; these cells secrete both IFNγ and IL-17 and thus share phenotypic characteristics of T<sub>H</sub>1 and T<sub>H</sub>17 cells, and also express the drug efflux protein p-glycoprotein, which might explain the resistance of T<sub>H</sub>17 cells to glucocorticoids in this context<sup>11,33</sup>. Circulating CD4<sup>+</sup> T cells and T<sub>H</sub>17

cells that express the exhaustion marker PD1 are also elevated in TAK. These cells drive fibrosis in other disease states, and might contribute to arterial-wall fibrosis in TAK, although these cell populations have not yet been studied in arterial tissue from patients with TAK<sup>11,18</sup>. Infiltrating CD8<sup>+</sup> cytotoxic T cells in TAK have increased T cell receptor activation and are important sources of IFNγ<sup>34,35</sup>.

The role of B cells in TAK is emerging. Tertiary lymphoid structures (TLSs) comprising follicular dendritic cells, memory B cells and T cells, with enrichment of antigen-experienced T follicular helper cells, are more prevalent in the arterial wall of TAK than GCA, particularly in active TAK<sup>16,17</sup>. In arterial tissue from patients with GCA and in a functional mouse model of GCA, stem-like CD4<sup>+</sup> T cells expressing TCF1 were crucial for TLS formation; however, this aspect remains unexplored in TAK<sup>36</sup>. Sera from patients with TAK can activate mTORC1 in endothelial cells in vitro<sup>37</sup>. Autoantibodies that target endothelial protein C receptor and scavenger receptor class B type I block these proteins from inhibiting endothelial-cell and T<sub>H</sub>17-cell activation, thereby driving inflammation in TAK<sup>38</sup>. Novel autoantibodies against proline-rich protein HaeIII subfamily 2, dihydropteridine reductase and spermatogenesis-associated 7 proteins were identified in 2021 using serum proteomics that distinguished samples from people with TAK from samples from healthy individuals, but these results require further validation<sup>39</sup>.

## Arterial-wall fibrosis

The inflamed TAK artery heals and becomes a fibrosed, stenosed artery. M2-like macrophages, mast cells, T cells (particularly PD1-expressing T<sub>H</sub>17 cells) and the cytokines IL-6 and IL-17A are postulated to drive fibroblast activation in the TAK artery. As arterial inflammation in TAK can result in fibrosis, using in vitro and preclinical models of disease to test if combining immunosuppressive and antifibrotic therapies can reduce subsequent arterial stenosis is important<sup>18</sup>. Several targeted therapies for TAK have been developed on the basis of the current understanding of disease pathogenesis<sup>32,40–46</sup> (Fig. 2).

## Animal models of Takayasu arteritis

Animal models that replicate the pathology of TAK are sparse. One model, which was first described in GCA and is now also used as a model of TAK, involves implanting human arteries from cadavers into immunodeficient NOD-SCID mice followed by infusion of peripheral-blood mononuclear cells from patients with active TAK, and results in immune-cell engraftment in the implanted artery<sup>32</sup>. Other models of TAK include aortitis induced by knocking out IRF4-binding protein, IL-1 receptor antagonist knockout mice or bioengineered human arteries engrafted with monocytes, macrophages and dendritic cells<sup>47,48</sup>.

Taken together, the interplay of environmental triggers, dysregulated innate and adaptive immune responses, and maladaptive tissue repair drives the vascular pathology of TAK, underscoring the importance of mechanistic insights for the development of targeted therapies.

## Genetics of Takayasu arteritis

The pathogenesis of TAK remains incompletely understood. Given the substantial heterogeneity observed between different patients, there might be multiple causal factors that contribute to disease pathogenesis. Evidence of a genetic contribution to the disease aetiology comes from differences between populations in the prevalence of TAK, familial case occurrence and the confirmed genetic associations within and outside of the HLA region<sup>6</sup>. Unbiased genome-wide association studies have



been performed in TAK that characterize the genetic landscape of the disease<sup>49–53</sup>. Currently, over 100 genetic susceptibility loci for TAK have been identified, including a number of loci established in multiple cohorts with a genome-wide association studies level of significance (Table 1).

Genetic susceptibility loci in Takayasu arteritis

The most well-established genetic association in TAK is within the HLA class I region. Specifically, *HLA-B\*52* has been consistently linked to increased disease susceptibility across different populations<sup>54–56</sup>. Furthermore, *HLA-B\*52* is associated with earlier disease onset and a more aggressive disease course, including increased rates of aortic regurgitation and left ventricular dysfunction in TAK<sup>55,57,58</sup>. Fine-mapping studies within the HLA region have led to the identification of an additional genetic association in the HLA class II region, specifically localized to *HLA-DQB1* and *HLA-DRB1* in TAK<sup>49</sup>. This association appears to be independent of *HLA-B\*52*, albeit much less robust. Interestingly, similar genetic associations within both HLA class I and class II regions have been reported in GCA<sup>59</sup>; however, in GCA, the genetic association is stronger in the HLA class II region. These findings suggest that the relative contribution of associations in HLA class I versus class II regions might influence if a patient predisposed to large-vessel vasculitis develops TAK or GCA.

Beyond the HLA region, multiple non-HLA loci have been implicated in TAK, primarily genes involved in immune-system regulation<sup>6</sup>. Among these, *IL12B* encodes the p40 subunit of IL-12 and IL-23, both of which have key roles in pro-inflammatory immune responses<sup>49,50</sup>. Another important cytokine-related gene, *IL6*, has been associated with TAK, with functional studies indicating that genetic variants in this region might alter the expression of pro-inflammatory mediators in monocytes and macrophages<sup>26,51</sup>.

Other associated loci include *FCGR2A* and *FCGR3A*, which encode Fc receptors involved in immune-complex clearance, and *ETS2*, a transcription factor that regulates macrophage-driven inflammation<sup>24,49,53</sup>. Additional susceptibility loci that have been identified include *DUSP22*, *PTK2B*, *KLHL33*, *RPS9/LILRB3*, *LILR3A*, *SVEP1*, *CFL2* and *VPS8*, many of which are involved in immune signalling, vascular biology and inflammatory responses<sup>51–53</sup>.

Functional implications of susceptibility variants

Similar to other immune-mediated diseases, the majority of genetic susceptibility loci in TAK are located in non-protein-coding regions and are instead found in regulatory regions that influence gene expression and chromatin structure<sup>6,53</sup>. Functional genomic analyses show that TAK-associated variants are enriched in enhancer and promoter regions, particularly in monocytes, suggesting their role in modulating immune responses<sup>53</sup>. Further enrichment of genetic susceptibility loci within regulatory chromatin regions in B cells also suggests a role for B cells in the pathogenesis of TAK<sup>53</sup>.

The disease-risk allele in a key disease-associated genetic variant within *IL6* suppresses the expression of *GPNMB*, an anti-inflammatory gene, through chromatin looping and recruitment of the MEF2–HDAC repressive complex in monocyte-derived macrophages<sup>26</sup>. Risk alleles in the disease-associated chr21q22 locus enhance the expression of *ETS2*, a transcription factor implicated in macrophage-mediated inflammation<sup>24,53</sup>. These findings highlight how non-coding genetic variation contributes to disease pathogenesis by altering immune-cell activity in TAK.

Transcription factor binding analysis has shown that TAK-associated variants are enriched in binding sites for STAT and RUNX family transcription factors, which are critical regulators of immune

responses<sup>6</sup>. This finding suggests that genetic variants associated with TAK might influence the activation of key inflammatory pathways and therefore provide novel potential therapeutic targets for TAK.

Genetic-risk score and population differences

A cumulative genetic-risk score (GRS) analysis has provided insights into the genetic burden of TAK across populations<sup>6</sup>. The highest genetic risk has been observed in East Asian populations, where TAK prevalence is highest. South Asian and African populations also exhibit a relatively high GRS, whereas European and admixed American populations are at a lower genetic risk. These findings suggest that genetic factors at least partly underlie the differences in TAK prevalence across populations. However, it should be noted that the use of a GRS alone in immune-mediated diseases in individual patients is limited, largely owing to the fact that GRSs are based mainly on common genetic variants.

Clinical implications

The identification of genetic risk loci in TAK has potential clinical applications, including biomarker development, risk stratification and personalized-medicine approaches. Moreover, genetic studies in TAK can identify promising new therapeutic targets; for example, the genetic association with *IL12B* suggests that ustekinumab might warrant consideration as a candidate for future clinical evaluation in TAK<sup>44,45,60</sup>. The IL-6 pathway is already a target of existing biologic therapies, such as tocilizumab, which is used in the treatment of TAK<sup>26,51</sup>. Evidence from functional genetic studies suggests that targeting ETS2-driven macrophage inflammation could offer novel therapeutic strategies in TAK and other chr21q22-associated diseases<sup>24</sup>. Finally, drug-repurposing analyses based on genetic-association studies

Table 1 | Genetic susceptibility loci in Takayasu arteritis

| Gene                                | Odds ratio | Loci    |
|-------------------------------------|------------|---------|
| <i>HLA-B</i> and <i>MICA</i>        | 3.28       | HLA     |
| <i>MUC21</i>                        | 2.71       | HLA     |
| <i>CCHCR1</i>                       | 2.44       | HLA     |
| <i>HLA-DQB1</i> and <i>HLA-DRB1</i> | 2.34       | HLA     |
| <i>IL6</i>                          | 2.07       | Non-HLA |
| <i>PTK2B</i>                        | 1.94       | Non-HLA |
| <i>chr13q21</i>                     | 2.02       | Non-HLA |
| <i>FCGR2A</i> and <i>FCGR3A</i>     | 1.81       | Non-HLA |
| <i>RPS9</i> and <i>LILRB3</i>       | 1.64       | Non-HLA |
| <i>CFL2</i>                         | 1.62       | Non-HLA |
| <i>chr21q22</i>                     | 1.56       | Non-HLA |
| <i>KLHL33</i>                       | 1.55       | Non-HLA |
| <i>IL12B</i>                        | 1.54       | Non-HLA |
| <i>LILR3A</i>                       | 1.53       | Non-HLA |
| <i>DUSP22</i>                       | 1.52       | Non-HLA |
| <i>VPS8</i>                         | 1.51       | Non-HLA |
| <i>SVEP1</i>                        | 1.45       | Non-HLA |
| <i>HLA-G</i>                        | 1.43       | HLA     |

Loci within and outside of the HLA region that have been associated with the susceptibility to Takayasu arteritis with a genome-wide association studies level of significance ( $P < 5 \times 10^{-9}$ ) are depicted<sup>6</sup>.

have identified medications such as fostamatinib and leflunomide as potential treatments for TAK<sup>61</sup>.

In summary, integrating genetic discoveries with functional genomics and clinical data will be essential for advancing understanding of the molecular mechanisms underlying TAK. Future research into the genetics of TAK should also focus on understanding gene–environment interactions in the development of TAK.

## Clinical features and differential diagnosis

The clinical presentation of patients with TAK depends on the stage of the disease and the distribution of vascular lesions. Classically, the natural history of TAK progresses through three stages: the first stage is characterized by the presence of non-specific constitutional symptoms (that is, non-specific symptoms that indicate generalized disease and do not signify specific organ pathology) caused by systemic inflammation; the second stage is vascular inflammation that presents with features such as carotidynia; the final stage is vascular insufficiency<sup>62</sup>. However, less than one-fifth of patients progress sequentially through such a triphasic disease<sup>63</sup>.

Most patients present with clinical features suggestive of underlying vascular inflammation and vascular insufficiency. The symptoms of vascular insufficiency, including limb claudication, absent pulses, vascular bruit and hypertension, are by far the most common presenting features, reported in more than 50% of patients with TAK<sup>64</sup>. Although some symptoms, including limb claudication, reflect the underlying vascular area of stenosis, many of the symptoms are multifactorial. For instance, breathlessness, cough and chest pain might be manifestations of multiple pathologies, including aortic regurgitation, cardiac failure, coronary heart disease, systemic hypertension and rarely pericarditis, pulmonary hypertension or aortic dissection<sup>65</sup>. A few patients might present with features of subclavian steal syndrome, which includes light-headedness, presyncope or syncope along with features of upper limb claudication<sup>66</sup>.

Around 30% of patients with severe narrowing of large arteries present with complications that include stroke or transient ischaemic attacks, hypertensive emergency, myocardial ischaemia, heart failure, aortic regurgitation, ocular blindness and, rarely, digital gangrene<sup>64,67,68</sup>. Patients can also present with features of left-ventricular dysfunction and pulmonary hypertension, including haemoptysis, but these features are rare<sup>69–71</sup>. Visual symptoms, such as amaurosis fugax and monocular or bilateral visual loss, are common and generally indicate advanced disease<sup>72</sup>. In many patients, extensive vascular damage is already present at the time of diagnosis, and life-threatening or organ-threatening complications might be the initial presenting features<sup>63,65,66,68,73</sup>.

In long-standing disease, the risk of cardiovascular disease, including stroke and transient ischaemic attack, is increased, with an estimated pooled prevalence of 15.8% (refs. 74–76). Coronary-artery lesions have also been described in TAK<sup>77</sup>; coronary-artery involvement occurs in ~5% to 50% of patients with adult-onset TAK, whereas the frequency in paediatric TAK varies from ~8% to 40% (ref. 78). For more than two thirds of patients, involvement occurs primarily in the proximal region and ostia of the coronary artery, whereas the middle and distal regions are rarely affected. Skip lesions (segmental areas of vascular inflammation separated by normal-appearing artery) are also characteristic of TAK-associated coronary-artery lesions<sup>79</sup>. Coronary-artery involvement is associated with a high frequency of major cardiovascular events, and this frequency is increased further with disease involving the coronary-artery ostia, ≥70% luminal stenosis and high disease activity<sup>79</sup>.

Multi-detector CT imaging is the preferred method for detecting coronary involvement<sup>80</sup>. Given the high frequency of coronary-artery lesions in TAK and the considerable impact of coronary lesions on cardiovascular mortality, it might be prudent to assess the coronary arteries of patients with TAK at least at diagnosis.

Carotidynia, a characteristic symptom of vascular inflammation, is described in 5% to 15% of patients with TAK. Up to one third of patients with TAK present with fatigue, fever, myalgia and malaise, which indicates systemic inflammation along with features of vascular inflammation or ischaemia. However, constitutional symptoms alone occur in less than 10% of patients<sup>62,64</sup>. The absence of ischaemic symptoms in these patients can lead to diagnostic delay<sup>81</sup>. Skin lesions, including erythema nodosum, pyoderma gangrenosum and cutaneous vasculitis, are uncommon and often indicate overlap with other autoimmune conditions<sup>82–84</sup>. A few (<10%) patients are asymptomatic and are diagnosed incidentally during evaluations for other medical conditions<sup>63</sup>. Some patients only present with vascular bruit (an abnormal sound heard over an artery caused by turbulent blood flow, usually owing to stenosis or other vascular irregularities), pulse asymmetry or blood-pressure abnormalities during examination.

The frequency of symptoms differs slightly with age at onset. Although constitutional symptoms, abdominal pain, hypertension, cardiomyopathy and renal failure are more common in childhood TAK, symptoms of peripheral ischaemia, including limb claudication, is more frequently observed in patients with adult-onset disease<sup>85</sup>. However, pulse abnormality and hypertension are still the commonest clinical presentations in children<sup>64</sup>. Infants most commonly present with hypertension, blood-pressure inequality between limbs and fever<sup>86</sup>. Aortic regurgitation and stroke are more frequently observed in adult TAK (approximately 7.5%) than in childhood-onset TAK<sup>64</sup>.

The co-occurrence of other immune-mediated diseases with TAK is not uncommon. The most common associations include spondyloarthritis (referred to as TAK-SPA) and inflammatory bowel disease, which are observed in approximately 6%–20% and 2%–6% of patients with TAK, respectively<sup>87</sup>. These conditions share common genetic susceptibility loci, including MHC class I and *IL12B*. The TAK-SPA group of patients present at a younger age and require biologic drugs more frequently than patients with TAK without SPA<sup>87</sup>.

## Differential diagnosis of Takayasu arteritis

Although imaging abnormalities in TAK are quite characteristic, many other autoimmune and non-autoimmune diseases might present with overlapping features and therefore need to be ruled out<sup>88</sup> (Table 2). The most common differential diagnosis is GCA. Older age at onset, more frequent cranial and constitutional symptoms (including polymyalgia rheumatica) and higher and persistent metabolic activity on PET scans, with less severe thickening of the involved areas, differentiate GCA from TAK<sup>89,90</sup>. In addition, patients with GCA generally do not present with lower-limb claudication and carotidynia<sup>88,91</sup>. Although the carotid and subclavian arteries can be involved in both diseases, longer segment and tapering stenotic lesions are more often found in GCA than in TAK<sup>92</sup>. IgG4-related peri-aortitis is another condition that can also mimic TAK<sup>93–95</sup> (Table 2). Atherosclerosis is yet another common mimic, particularly in older individuals; in contrast to the smooth long segment concentric mural thickening in TAK, atherosclerosis is characterized by eccentric thickening of the aorta or its branches with focal, low-intensity uptake in fluorodeoxyglucose (FDG)-PET studies<sup>96</sup>. Atherosclerosis preferentially involves the distal portions of the carotids and its branches<sup>96</sup>. Some infections can also resemble TAK; tuberculosis

**Table 2 | Differential diagnosis of Takayasu arteritis**

| Disease(s) or disorder(s)                                                 | Prominent features                                                                                                                                                 | Vascular imaging findings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Refs.   |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>Autoimmune and inflammatory diseases</b>                               |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Giant-cell arteritis                                                      | Older age at disease onset than in TAK<br>Frequent constitutional and cranial symptoms                                                                             | Superficial temporal arteries and cranial and axillary arteries are commonly affected<br>PET-CT scans show higher and persistent metabolic activity with less severe mural thickening of the involved arteries than in TAK                                                                                                                                                                                                                                                                                    | 89,90   |
| IgG4-related peri-aortitis                                                | Other organ involvement<br>The presence of retroperitoneal fibrosis                                                                                                | The aorta and coronary arteries are most affected, followed by iliac arteries<br>Diffuse or patchy mural thickening of often >2mm along with homogenous delayed wall enhancement ('mantle sign') with either aneurysms or stenosis<br>Periaortic soft-tissue thickening and absence of mural calcifications differentiate IgG4-related peri-aortitis from TAK<br>FDG-PET scans show no or minimal metabolic activity in the aortic wall but metabolic activity is present in peri-aortic and arterial tissues | 93–95   |
| Associated with other autoimmune rheumatic conditions <sup>a</sup>        | Disease-specific features that are indicative of other organ involvement and the presence of autoantibodies specific for other rheumatic diseases                  | Small to large vessel involvement<br>Behçet disease might lead to both aneurysms and stenosis with involvement of the pulmonary arteries, the aorta and aortic branches<br>Ascending aorta involvement with aortic insufficiency is a common feature of Cogan's syndrome                                                                                                                                                                                                                                      | 88,89   |
| Antiphospholipid antibody-associated vasculopathy                         | Previous pregnancy loss or pregnancy complications<br>Vascular thrombosis<br>Associated with neurological involvement<br>Normal inflammatory markers               | Involvement of any of the aortic-arch branches, including carotids with the internal carotid artery, innominate and subclavian arteries<br>Usually non-inflammatory with no or minimal metabolic activity                                                                                                                                                                                                                                                                                                     | 188     |
| Erdheim–Chester disease                                                   | Clinical features that reflect the involvement of other organs, including the lungs, eyes, long bones, skin, pituitary gland and brain<br>Retroperitoneal fibrosis | Circumferential wall thickening of the aorta (coated aorta), the entire aorta is involved in 50% of patients<br>Descending thoracic aorta and renal ostial stenosis<br>Periarteritis is common                                                                                                                                                                                                                                                                                                                | 189–191 |
| <b>Non-infectious non-autoimmune diseases</b>                             |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Atherosclerosis                                                           | Affects older patients<br>Absence of symptoms such as carotidynia<br>Normal or mildly elevated inflammatory markers                                                | The disease might extend up to and beyond the carotid bifurcation, which are usually spared in TAK<br>Subclavian arteries are less commonly involved<br>The aortic-wall thickening is focal, eccentric and irregular in atherosclerosis, whereas smooth, concentric and long-segment thickening is the typical involvement in TAK<br>Calcified plaques on CT angiography are more common than in TAK<br>FDG uptake is less intense, with an SUV max of <2                                                     | 96      |
| Fibromuscular dysplasia                                                   | Primarily affects young women<br>Usually presents with hypertension or mid-aortic syndrome, stroke<br>Normal inflammatory markers                                  | Typically characterized by symmetrical involvement of bilateral renal and distal carotid arteries, predominantly the extra-cranial portion of the internal carotid artery<br>Characteristic 'string of beads' appearance, owing to areas of focal stenosis and aneurysms                                                                                                                                                                                                                                      | 192     |
| Segmental mediolysis                                                      | Abdominal pain is a frequent symptom                                                                                                                               | Affects medium-sized arteries, including involvement of the superior mesenteric and coeliac axis, whereas the main aorta is usually spared<br>Characterized by non-inflammatory dissecting aneurysms<br>Aneurysms and dissecting haematoma lesions can also be observed                                                                                                                                                                                                                                       | 193,194 |
| <b>Hereditary elastic fibrous disorders</b>                               |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Vascular Ehlers–Danlos syndrome, Loeys–Dietz syndrome and Marfan syndrome | Positive family history<br>Characteristic clinical features, including tall stature, ectopia lentis and hyperextensibility of skin and joints                      | Involvement of the infra-diaphragmatic aorta and infra-diaphragmatic aorta branches<br>Multiple arterial dissections and aneurysms, and rarely occlusions<br>No or low metabolic activity on PET scans                                                                                                                                                                                                                                                                                                        | 195     |
| Neurofibromatosis-1 vasculopathy                                          | Characteristic features, including symptoms suggestive of compression of the gastrointestinal tract, pulmonary and urinary tracts by neurofibroma                  | Arterial aneurysms of the aorta, renal, carotid or mesenteric arteries with or without stenosis<br>Arterio-venous malformation                                                                                                                                                                                                                                                                                                                                                                                | 196     |

Table 2 (continued) | Differential diagnosis of Takayasu arteritis

| Disease(s) or disorder(s)                                                        | Prominent features                                                                                                                                | Vascular imaging findings                                                                                                                                                                                                                                                                                                                                                                                                   | Refs.     |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Infectious aortitis disorders</b>                                             |                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |
| Mycobacterium tuberculosis arteritis, syphilitic aortitis and bacterial aortitis | These disorders typically present with non-specific constitutional symptoms, including fever, weight loss, malaise and myalgia<br>Lymphadenopathy | Abdominal aorta is primarily involved, followed by descending thoracic aorta and renal arteries<br>Multiple saccular aneurysms without substantial wall thickening can be observed in tuberculous arteritis<br>Arterial-wall erosions might occur (uncommon in TAK)<br>Non-homogenous contrast-medium uptake, perivascular inflammation with perilesional lymphadenopathy<br>FDG-PET uptake in large arteries can be patchy | 97,98,197 |
| <b>Miscellaneous diseases involving a single large vessel</b>                    |                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                             |           |
| Transient perivascular inflammation of the carotid artery syndrome               | Acute neck pain with transient neurological symptoms                                                                                              | Only carotid abnormality with peri-arterial infiltration with focal soft plaque (30% of people) and luminal narrowing (15% of people)<br>All findings resolve or are substantially reduced within days or months                                                                                                                                                                                                            | 198       |
| Thoracic outlet syndrome                                                         | Symptoms related to the vascular area involved                                                                                                    | Typically affects the mid-distal segment of the subclavian artery                                                                                                                                                                                                                                                                                                                                                           |           |
| Congenital hypoplasia of the carotid or subclavian arteries                      | Symptoms related to the vascular area involved                                                                                                    | Usually presents as isolated long-segment stenosis or 'string appearance'                                                                                                                                                                                                                                                                                                                                                   | 199       |
| Post-radiation carotid stenosis                                                  | History of radiation with only carotid involvement                                                                                                | Carotid involvement                                                                                                                                                                                                                                                                                                                                                                                                         | 200       |

\*Including Cogan's syndrome, relapsing polychondritis, anti-neutrophil cytoplasm antibody-associated vasculitis, sarcoidosis, Behçet disease, axial spondyloarthritis and Sjögren disease. SUV, standard uptake value; TAK, Takayasu arteritis; FDG, fluorodeoxyglucose.

is the most common mimic of TAK, especially in developing countries<sup>97</sup>. The presence of constitutional symptoms and multiple aneurysms on imaging studies can indicate possible tuberculous arteritis<sup>97,98</sup>. In patients with predominantly aneurysmal lesions, inherited disorders of connective tissue, such as Ehlers–Danlos syndrome or Loews–Dietz syndrome, merit consideration. Many other differential diagnoses exist for TAK, and knowledge of the clinical and imaging features of each might prevent misdiagnosis (Table 2).

Imaging in the diagnosis and management of Takayasu arteritis

Imaging has a central role in the diagnosis, classification and management of TAK. Advances in imaging technology have transformed how clinicians detect vascular inflammation, monitor disease progression and assess response to treatment. The following section highlights the evolving role of vascular imaging in TAK.

Importance of imaging

In 2022, the classification criteria for TAK were updated<sup>99</sup>. Compared with the original 1990 classification criteria for TAK<sup>100</sup>, the updated criteria more strongly reflect the increasing importance of vascular imaging to define and characterize TAK. An absolute requirement of the updated criteria is that, to be classified as TAK, a patient must have imaging evidence suggestive of vasculitis in the aorta or branch arteries. Although the criteria permit confirmation of vasculitis by many different imaging techniques (angiography, PET or ultrasonography), this mandated requirement emphasizes the role that imaging has as a complementary assessment to the vascular physical examination in TAK<sup>101</sup>.

Vascular-imaging modalities

Multiple imaging techniques can be used to assess the large arteries. Non-invasive angiography by MRI or CT has largely replaced catheter-based angiography, except for instances in which concomitant direct measurement of central arteries is desirable or during

endovascular interventions. PET imaging can be used to assess arterial <sup>18</sup>F-FDG uptake as a surrogate measure for vascular inflammation, typically coupled with either CT- or MR-based anatomical imaging<sup>102,103</sup>. Ultrasonography is an inexpensive way of assessing parts of the vasculature at the bedside. Each modality has specific advantages and limitations; consequently, multimodal imaging can be used to assess patients comprehensively, particularly when there are unexpected findings by one form of imaging assessment<sup>104</sup>. Consensus-based recommendations for the use of imaging in large-vessel vasculitis have been proposed to standardize image acquisition and interpretation<sup>105,106</sup>.

The purpose of vascular imaging in TAK is to identify luminal disease (for example, aneurysm, stenosis or occlusion) and to characterize morphological abnormalities of the vascular wall (such as increased wall thickness or oedema). Differentiating vascular damage from active disease can be challenging but has important treatment implications<sup>107</sup>. Increased wall thickness might reflect ongoing vascular inflammation, prior damage or vascular remodelling, whereas vascular oedema on MR angiography or increased arterial FDG uptake by PET is more suggestive of active vasculitis<sup>108</sup>. Technological advances in imaging will continue to improve the ability to monitor disease at the vascular level. Specific MRI sequences aid the visualization of wall thickness (such as black blood imaging) or oedema (such as signal tau inversion recovery). Novel radioligands that target specific immune-cell populations, such as fibroblast-activating protein inhibitor for fibroblast activation and somatostatin 2 receptor to indicate macrophage infiltration, might improve the specificity of PET by detecting and defining inflammation within the vascular wall compared with FDG<sup>109</sup>. Newer-generation PET scanners will enable faster imaging with less radiation and define the kinetics of radiotracer distribution within the vascular compartment<sup>110</sup>.

Angiographic patterns

Various classification patterns have been proposed to define subgroups of patients with TAK with different angiographic patterns of disease. The Lupi-Herrera classification scheme, a modification of



the Ueno classification<sup>111</sup>, divides patients into four subtypes on the basis of involvement of the aortic-arch branch vessels, descending thoracic, abdominal, and pulmonary arteries<sup>62</sup>. The Numano classification groups patients into five subtypes on the basis of involvement of the aortic arch, aortic-arch branch vessels, ascending aorta, descending thoracic aorta, abdominal aorta and renal arteries<sup>112</sup>.

Data-driven classification patterns were developed by assessing differences in angiographic patterns of damage from over 800 patients with TAK from India and North America. Three clusters were identified: involvement of the abdominal aorta and its branches; involvement of the aortic arch and its branches; and asymmetric, focal disease<sup>113</sup> (Fig. 3). Angiographic patterns differ among patients with TAK in different parts of the world. Patients in India have more frequent abdominal vasculature involvement, whereas patients from Japan and North America are more likely to have involvement of the aortic arch and its branches<sup>112–114</sup>. Ongoing studies are assessing the extent to which genetics might influence angiographic patterns of disease.

## Imaging to monitor disease activity

Discordance between clinical symptoms and imaging findings can occur when assessing disease activity in patients with TAK. New or worsening areas of arterial damage have been noted on serial angiography in patients with TAK during periods of apparent clinical remission<sup>115</sup>. Given the concern for ‘silent’ angiographic progression of disease, some guidelines recommend that patients with TAK undergo regularly scheduled non-invasive angiography to assess for changes in vascular damage over time, even in the absence of clinical symptoms of disease activity; however, the optimal timing for repeat imaging is not specified owing to limited evidence<sup>116</sup>. The development of a new lesion in a previously unaffected vascular area is of greater concern, as it indicates active disease, compared with worsening of existing arterial stenosis, which can be secondary to changes in scar tissue within the arterial wall (that is, ‘healing fibrosis’)<sup>116</sup>.

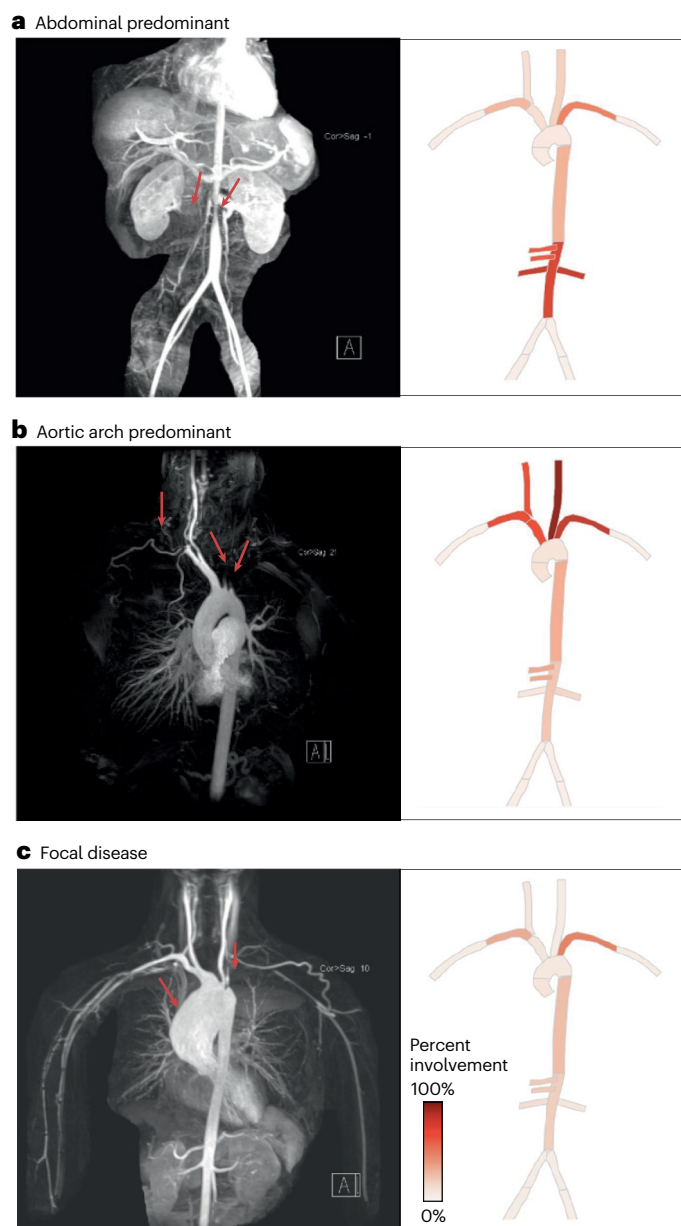
Changes in the vessel wall on imaging that might suggest active disease include vascular oedema, contrast enhancement and increased wall thickening on CT or MR imaging, or vascular FDG uptake in the arterial wall on FDG-PET. These imaging sequences can be complementary to clinical assessment in TAK, and particularly helpful for instances in which there is clinical uncertainty about disease activity; however, these findings are not always specific indicators of active disease, and other factors, such as vascular remodelling and atherosclerosis, can contribute to wall-morphology changes and vascular FDG uptake<sup>106,108,117,118</sup>. FDG uptake in the arterial wall better predicts future stenotic lesions in TAK than wall thickness alone<sup>119</sup>, but the majority of arterial regions with FDG uptake do not develop stenosis (>90%)<sup>117</sup>. Arteries with wall thickening and oedema on MRI and concomitant FDG uptake on PET are at a high risk of angiographic progression<sup>117</sup>. In the absence of prospective studies to inform specific guidelines regarding serial imaging to monitor disease activity, these decisions should be individualized to each practitioner in alignment with institutional resources and capabilities.

Scoring systems have been proposed to quantify changes in the vessel wall on imaging over time (such as the PET Vascular Activity Score<sup>102</sup> or Vasculitis Activity using MRI and PET scores<sup>120</sup>), but these have mainly been used in research settings and are not routinely incorporated into clinical practice. Novel disease indices that integrate clinical and imaging assessments in TAK have also been proposed, including the TAK Integrated Disease Activity Index. This index evaluates clinical, imaging and laboratory findings in a hierarchical manner, and vascular FDG uptake on PET is only scored if a patient has a corresponding

disease-associated clinical symptom<sup>121</sup>. Incorporating vascular imaging as an outcome measure will probably occur with increasing frequency in future therapeutic trials, and these data will be important to determine the extent to which serial imaging is helpful in assessing response to therapy and guiding management.

## Biomarkers and disease-activity monitoring

Identifying accurate and reliable biomarkers to assess disease activity in TAK remains challenging. Traditional inflammatory markers, such



**Fig. 3 | Angiographic patterns of Takayasu arteritis.** Using k-means clustering, three angiographic clusters have been identified and validated in TAK: **a**, abdominal predominant; **b**, aortic-arch predominant; **c**, focal disease. Representative angiograms of a patient in each cluster are depicted with sites of vascular damage identified by red arrows. Heatmaps of the large arteries indicate the percentage of patients in each cluster with involvement of each artery. Adapted with permission from ref. 203, Wiley.

## Box 1 | Biomarkers of active disease in Takayasu arteritis

### Acute-phase reactants

- Erythrocyte sedimentation rate
- C-reactive protein
- Pentraxin-3
- Serum amyloid A
- Complement C4-binding protein
- Complement 3
- Complement 1q

- Monocyte chemotactic protein-1
- CCL2
- CCL20
- CCL22
- CXCL8
- CXCL10

### Matrix metalloproteinases (MMP)

- MMP-3
- MMP-9

### Cytokines

- IL-6
- IL-18
- IL-2
- IL-17A
- Soluble IL-6 receptor
- IFN $\gamma$

### Autoantibodies

- Anti-endothelial cell antibody
- Anti-cardiolipin antibody
- Anti-annexin V antibody

### Endothelial-damage markers

- Circulating endothelial progenitor cell
- Vascular endothelial growth factor
- Endothelin-1

### Others

- T helper (T<sub>H</sub>) 1 cells
- T<sub>H</sub>17 cells and T<sub>H</sub>17.1 cells
- Macrophage inflammatory protein-1 $\beta$
- Leucine-rich  $\alpha$ 2-glycoprotein
- Platelet-to-lymphocyte ratio
- Neutrophil-to-lymphocyte ratio
- Plateletcrit

### Chemokines

- CCL5 (also known as RANTES)

as erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP), are widely used<sup>122</sup>, but lack sufficient sensitivity and specificity<sup>123,124</sup>. Active disease can be present in up to one third of patients with normal ESR and CRP levels<sup>53,125</sup>. Moreover, ESR levels can be high in 44% of patients who have achieved remission<sup>126</sup>. When the levels of different acute-phase proteins in patients with TAK were assessed, only serum amyloid A (SAA) and complement C4-binding protein (C4BP) levels were increased in patients with active TAK compared with patients with inactive disease and healthy individuals<sup>127</sup>. Another study confirmed that SAA levels were higher in patients with active disease than in those with stable disease and decreased after treatment<sup>128</sup>. These findings suggest that SAA might be a reliable biomarker for assessing disease activity and response to treatment in TAK.

Several cytokines, chemokines, autoantibodies and growth factors are increased in TAK, but they are not disease specific<sup>129</sup>. Among these markers, IL-6, IL-18, CCL5 (also known as RANTES), monocyte chemotactic protein-1, CCL2, CCL20, CCL22, CXCL8, CXCL10, matrix metalloproteinase-3, matrix metalloproteinase-9, anti-endothelial cell antibody, anti-cardiolipin antibody, anti-annexin V antibody, serum C1q levels and leucine-rich  $\alpha$ 2 glycoprotein were associated with clinical disease activity<sup>27,28,130–138</sup> (Box 1). Higher complement protein C3 levels occur in patients with active disease than in those with inactive disease (sensitivity of 69.9%, specificity of 66.7%) and can better indicate

active TAK than IL-6 levels<sup>139</sup>. However, all these biomarkers need to be investigated further before they can be applied in clinical settings.

A 2024 systematic review and meta-analysis reported that IL-6 levels were increased in patients with active TAK compared with those with inactive disease and healthy individuals<sup>140</sup>. In another study, soluble IL-6 receptor levels were also increased in patients with active but not inactive TAK compared with healthy individuals and could be a potential biomarker for disease activity<sup>141</sup>. Prospective longitudinal studies are needed to assess the clinical use of IL-6 as a biomarker of disease activity in TAK.

The pentraxin (PTX) superfamily is a group of proteins that function as acute-phase reactants during exogenous pathogen recognition. PTX-3 is a long pentraxin that is secreted independently of IL-6 stimulation and has been proposed as a potential marker for active disease in TAK<sup>142–147</sup>. However, two large studies found no difference in PTX-3 levels between patients with active and inactive disease<sup>148,149</sup>. By contrast, another study reported elevated PTX-3 levels in a subgroup of patients with TAK who showed signs of active inflammation on vascular imaging<sup>149</sup>. In addition, PTX-3 levels correlate strongly with disease activity, as assessed by PET imaging and standardized indices such as the NIH criteria and ITAS2010 (ref. 150). Furthermore, PTX-3 levels show stronger correlation with activity scores in PET scans and ITAS2010 than ESR and CRP<sup>151</sup>. In addition, results of a systematic review and meta-analysis of eight studies indicate that PTX-3 might be more valuable than CRP for the assessment of disease activity in TAK<sup>152</sup>; however, this result should be interpreted cautiously owing to the heterogeneity and potential publication bias of this meta-analysis. Furthermore, although PTX-3 seems to be a promising biomarker for disease activity in TAK, its use in clinical practice and correlation with disease activity in imaging studies need to be evaluated in prospective follow-up studies.

Thus, reliable biomarkers for assessing active disease in TAK remain limited, largely owing to conflicting findings across studies; however, PTX-3 seems to be the most promising biomarker for monitoring disease activity in TAK. CRP, ESR and SAA are still the most commonly used biomarkers to assess disease activity in TAK, but it should be noted that these biomarkers are non-specific and might not accurately reflect disease activity. Moreover, they are usually suppressed with IL-6-blocking agents, which are commonly used in the management of TAK.

Identifying biomarkers of arterial-wall fibrosis and damage in TAK is an area of increasing interest. Higher Enhanced Liver Fibrosis scores (used to assess fibrosis in liver diseases) and TAK-associated polymorphisms in *IL12B* have been shown to predict more vascular damage in patients with TAK<sup>153,154</sup>.

## Management of Takayasu arteritis

Effective management of TAK requires a carefully balanced approach to control inflammation while minimizing treatment-related risks. This section reviews current therapeutic strategies, including the use of glucocorticoids, conventional immunosuppressive agents and biologic therapies, highlighting evolving recommendations in TAK treatment (Fig. 4).

Glucocorticoids are the mainstay treatment for remission induction in TAK<sup>155</sup>. However, glucocorticoid monotherapy for remission induction is associated with frequent relapses after drug withdrawal and is not recommended in current management guidelines<sup>116,156–160</sup>. Most guidelines suggest doses of 0.5–1 mg/kg per day (40–80 mg per day) prednisone or prednisolone as the initial therapy<sup>116,156–159,161,162</sup>, except for

milder, single lesions, for which lower doses 25–30 mg per day might be used<sup>159,162</sup>. Similar complete remission and relapse rates were shown in patients receiving 0.5 mg/kg per day and 1 mg/kg per day as the initial glucocorticoid dose when a second immunosuppressive agent is used<sup>163</sup>. No studies have compared the efficacy of high-dose glucocorticoid versus pulse glucocorticoid treatments for TAK; however, pulse glucocorticoids are suggested for refractory, life-threatening or organ-threatening disease<sup>116,160,161</sup>. Although there is no standard glucocorticoid dose-reduction regimen for TAK, most guidelines recommend lowering prednisone doses to <10 mg per day, and some guidelines from the past 3 years recommend aiming for <5 mg per day after 1 year<sup>116,156,157,159–162</sup>. Low-dose or discontinuation of glucocorticoid monotherapy, especially in the first year, led to relapses in 60–77% of patients during two randomized controlled trials (RCTs)<sup>40,43</sup>.

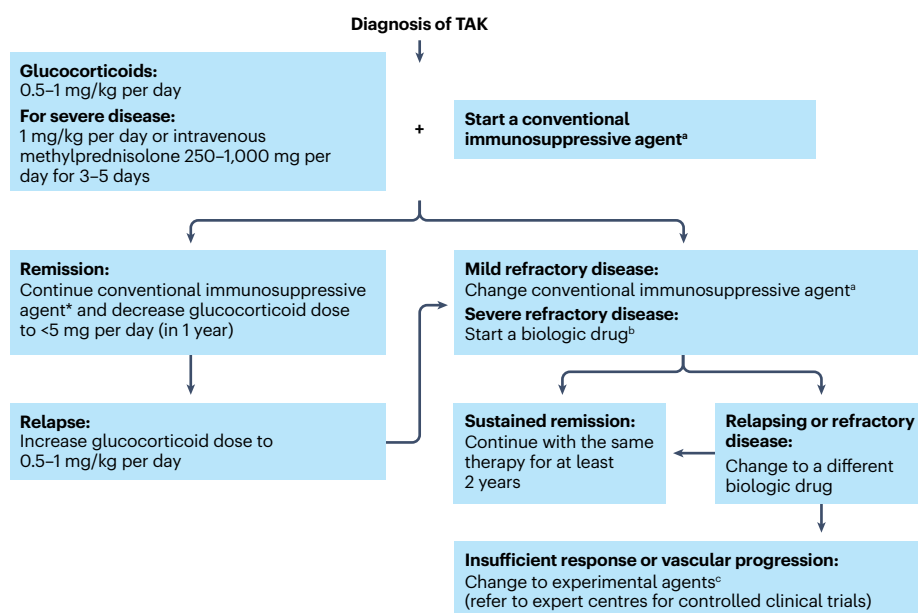
Owing to the toxicity risk that can occur with long-term glucocorticoid use and high relapse rates with glucocorticoid monotherapy, additional immunosuppressive agents are recommended from the initial remission-induction phase<sup>116,156,157,159,160</sup>. Efficacy of methotrexate, azathioprine, leflunomide and mycophenolate mofetil on induction of remission and prevention of relapses were shown in various observational studies and meta-analysis<sup>164–166</sup>. Methotrexate and azathioprine have a similar efficacy, with low long-term retention rates as first-line immunosuppressive drugs<sup>167</sup>. Mycophenolate mofetil and methotrexate combination therapy was shown to be superior to cyclophosphamide and azathioprine sequential therapy<sup>168</sup>. Owing to the serious adverse effects of cyclophosphamide and the lack of the proven superiority of this drug against other conventional immunosuppressive drugs, cyclophosphamide is only recommended for severe life- or organ-threatening disease<sup>157,162,169</sup>.

Clinical or angiographically refractory disease to conventional immunosuppressive drugs or frequent flares are a clinical challenge in the management of TAK. Biologic therapies are recommended in these patients in all current guidelines; however, using biologic drugs from the onset in patients with a severe disease course is also recommended in some guidelines<sup>116,157,162</sup>. The efficacy of TNF inhibitors for remission induction and prevention of relapses in TAK was demonstrated in

multiple observational studies<sup>170</sup>. Retrospective, comparative data indicate that using TNF inhibitors can lead to a better disease course, without new vascular lesions, than glucocorticoid monotherapy and immunosuppressive agents<sup>171</sup>. Despite an RCT of tocilizumab (an anti-IL-6 receptor antagonist) versus placebo failing to meet its primary end point of time to first relapse<sup>40</sup>, the efficacy of tocilizumab has been shown in numerous observational studies<sup>41</sup>. There are only a limited number of retrospective studies that compare TNF inhibitors and tocilizumab, with no clear evidence of superiority between them<sup>172,173</sup>. However, one prospective, comparative trial demonstrated better outcomes with adalimumab (a TNF inhibitor) at 24 weeks<sup>174</sup>. Although most recommendations do not make a preferential choice between the two classes of biologic drugs<sup>156–159,162</sup>, the American College of Rheumatology guidelines recommend TNF inhibitors owing to greater clinical experience and more available data, and the challenge of using acute-phase reactants to monitor disease activity in TAK when treated with tocilizumab. Unlike TNF inhibitors, tocilizumab can inhibit the acute-phase response (such as C-reactive protein production), even in the presence of clinically active disease<sup>116</sup>.

For refractory disease or patients who are intolerant to TNF inhibitors or tocilizumab, new alternative agents are emerging. Secukinumab (an IL-17 inhibitor) was shown to have similar efficacy to TNF inhibitors in a small prospective trial<sup>42</sup>. Among Janus kinase (JAK) inhibitors, tofacitinib has the most comprehensive data, favouring its use over methotrexate but not over leflunomide for TAK<sup>175,176</sup>. A prospective case series with another JAK inhibitor, baricitinib, showed favourable results<sup>177</sup>. Abatacept has shown negative results, and data on rituximab are limited; therefore, the use of these drugs is not currently recommended for TAK<sup>43,157</sup>. Data for ustekinumab in a 2025 RCT were inconclusive, as patient numbers were too low, owing to recruitment challenges<sup>44,45</sup>.

The optimal duration of immunosuppressive therapy for TAK is unknown<sup>157</sup>. Use of immunosuppressive agents, lower cumulative glucocorticoid dose and earlier disease onset are associated with reduced long-term disease-associated and treatment-associated damage in TAK<sup>178</sup>. Clinicians should be aware of risks associated with long-term glucocorticoid therapy and discontinuation of glucocorticoids should



**Fig. 4 | Therapeutic approaches to Takayasu arteritis.** Current treatment strategies for Takayasu arteritis (TAK) include glucocorticoids, conventional immunosuppressive agents and biologic therapies, which reflects the evolving recommendations in TAK management. \*Methotrexate, azathioprine, leflunomide, mycophenolate mofetil or cyclophosphamide; †TNF inhibitors or tocilizumab; ‡Janus kinase inhibitors, secukinumab, calcineurin inhibitors, sirolimus or rituximab.



be considered in patients with sustained remission for longer than 6–12 months. Addition of immunosuppressive agents, especially biologic agents, might decrease the relapse rates to less than 10% and enable glucocorticoid doses to be reduced to <10 mg per day<sup>179</sup>. In our opinion, immunosuppressive agents should be used for a minimum of 2 years after remission is achieved. Tapering of immunosuppressive drugs should also be slow, especially in patients with frequent relapses and organ- or life-threatening manifestations. Whether routine imaging should be used before immunosuppressive agent tapering or drug discontinuation is debatable, with no high-quality evidence. Evidence to support the routine use of low-dose aspirin and statins in all patients with TAK for primary cardiovascular prevention is insufficient. In our opinion, the use of these treatments should be limited to patients with conventional cardiovascular risk factors; however, in those with high-risk arterial involvement, such as coronary or cerebral arteries, these therapies might be considered.

Vascular interventions are considered mainly for patients with haemodynamically severe vascular stenosis or occlusions, such as cranial arterial or coronary disease, renal-artery stenosis or aortic aneurysms with a rupture risk, especially those with a progressive disease course, despite effective treatment with immunosuppressive drugs<sup>162</sup>. However, the American College of Rheumatology guidelines emphasize a more conservative approach, advocating for the escalation of immunosuppressive drugs or biologic therapies in cases of progressive vascular disease before considering vascular interventions<sup>116</sup>. Patients undergoing vascular interventions should be on lifelong antiplatelet agents<sup>180</sup>.

Vascular interventions should be planned, if possible, during clinical remission and with adequate immunosuppressive agents or biologic therapy before the procedure. In emergency situations, vascular interventions should be performed under the protection of a perioperative high-dose glucocorticoid therapy to prevent postoperative complications such as stenosis or occlusions<sup>160,181</sup>. However, risk of infection and possible impairment of wound healing with high-dose glucocorticoids should be assessed for each individual patient. The type of vascular intervention (open surgery versus endovascular procedure) in TAK can vary depending on the affected vascular region<sup>160</sup>. Open surgeries, such as bypasses, have a high postoperative rate of patency and are preferred in complicated and long-segment, occlusive lesions<sup>160</sup>. However, the risk of early perioperative mortality and complications are high with open surgeries. Endovascular therapies such as balloon angioplasty and stent implantation are minimally invasive, effective in short-segment lesions, but their long-term restenosis rates are higher<sup>159</sup>. However, repeat percutaneous interventions are possible, and patency has been shown to increase from 49% to 83% in obstructive lesions in a large study from India<sup>182</sup>. For patients with renal-artery involvement, endovascular intervention should be prioritized over open surgery, with balloon angioplasty preferred to stenting owing to its lower restenosis rate<sup>159,160,183,184</sup>.

Given the risk of complications such as stroke and death, endovascular intervention should also be the primary option for patients with supra-aortic artery involvement. For coronary-artery involvement, open surgery seems generally more successful than percutaneous coronary interventions and might be the preferred option<sup>185</sup>. However, the retrospective nature of these data and possible selection bias for patients with severe disease for endovascular interventions limit the interpretation of these observations. In patients with stable upper-extremity claudication, vascular interventions are generally not preferred, as the risk of critical ischaemia is low owing to extensive collateral circulation<sup>116</sup>.

Patients with TAK treated with glucocorticoids and immunosuppressive agents are at a greater risk of osteoporosis, infections and cardiovascular events. Such comorbid conditions should be managed as per standard recommendations<sup>186,187</sup>. Hypertension is another important manifestation of TAK that requires management. The presence of anatomical or functional (stenosis of suprarenal abdominal aorta or descending thoracic aorta) bilateral renal-artery stenosis in a substantial proportion of people with TAK precludes the use of angiotensin-converting enzyme inhibitors or angiotensin receptor blockers for managing hypertension<sup>180</sup>.

In summary, although additional evidence from clinical trials is needed, current data suggest that early and aggressive use of glucocorticoids with immunosuppressive or biologic agents results in better outcomes for patients with TAK, with reduced morbidity and mortality.

## Conclusion

Perturbations in innate and adaptive immune mechanisms, which are driven at least in part by genetic factors, promote the pan-arteritis observed in TAK. Extracranial giant-cell arteritis and atherosclerosis are important differential diagnoses, particularly in older individuals. Imaging is the cornerstone for diagnosing TAK and can help to delineate luminal narrowing and wall thickening. <sup>18</sup>F-FDG uptake in the arterial wall is increasingly recognized to denote active disease and arterial segments at a risk of future stenosis. Despite a lack of quality evidence from RCTs, glucocorticoids, along with conventional, biologic or targeted synthetic immunosuppressive agents, remain the cornerstone of the management of TAK.

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## Author contributions

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## Competing interests

The authors declare no competing interests.

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